

An Approach to EEG based BCI for Motor Imagery using Explainable Transfer Learning

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Abstract—In this contribution we investigate possibilities of using neural network classification methods for EEG based BCI. We focus on image-based feature representation. The time-frequency based representation and its selected variants along with bi-spectrum were used. Set of more experiments including performance comparison with KNN and more feature mappings is provided. The explainability analysis for the neural networks approach is provided along with corresponding findings.

Index Terms—Brain-computer interface, Motor imagery, Bispectrum, Explainable neural networks, Transfer learning

I. INTRODUCTION

Brain-Computer Interface (BCI) represent a remarkable innovation in the field of neuroscience, enabling direct communication between the human brain and external devices, such as computers or prosthetic limbs. BCI find extensive applications in various fields of bioengineering and neuroprosthetics. The most prevalent BCI rely on electroencephalogram (EEG) due to its noninvasive characteristics and cost-effective recording equipment, enabling seamless real-time functionality [1]. EEG based BCI is extensively utilized in conjunction with Motor Imagery (MI). MI refers to a cognitive procedure in which the individual envisions motion without engaging the muscles. Sensorimotor rhythms (SMR) arise within somatic and sensorimotor regions, predominantly within the Alpha (8-13Hz), Beta (13-30Hz), and Gamma (above 31Hz) frequency ranges. When specific body parts such as the legs are executed or envisioned in motion, it elicits a distinct SMR response [2]. A typical BCI system consists of an initial part in which we obtain a signal, followed by a signal processing, feature extraction and data classification. Currently, methods that use time-frequency methods for feature extraction in combination with deep neural networks are dominant in research. The authors in [2] used the STFT method. Spectrograms were created using electrodes C3, CZ and C4. They used the created spectrograms as input to CNN. They used several types of CNNs such as AlexNet, LeNet or EEGNet and achieved accuracy up to 94.41%. In the article [3], the authors also used the STFT method to which they added the CWT method for feature extraction. For classification they used AlexNet network. Using the CWT method, they achieved a success rate of up to 99.35%.

In our work, we will expand the research in [4] and we will focus primarily on time-frequency methods in connection with convolutional neural networks. To compare the convolutional network, we will use basic classifiers such as KNN and SVM. Since pre-trained models of neural networks are mostly used, we cannot determine on the basis of which parameters the pre-trained neural network makes decisions. There are several explainable methods that can determine on what basis the given network is trained and makes decisions. In our work, we will focus on these methods.

In parts II-V we present details about data choice and preparation, feature extraction, classification, and evaluation approach. In parts VI-VII we present the experiments, discuss their results and conclude.

II. DATASET CHOICE AND DATA PREPARATION

A. Dataset I.

In the BCI competition II. (2003) was published Dataset III, which is given by the Department of Medical Informatics, Institute for Biomedical Engineering, University of Technology Graz (Gert Pfurtscheller) [5]. The experiment was performed on a 25-year-old woman who was sitting in a chair. The woman's task was to control the movement of the rod by thinking about the movement of her right or left hand. Three bipolar EEG channels were measured on three electrodes - C3, CZ and C4. The EEG signal was sampled with a sampling frequency (fvz) of 128 Hz and was filtered between 0.5 and 30 Hz.

B. Dataset II.

In the online database GigaDB (GigaScience DataBase), was published a dataset focused on the representation of the movement of the right and left hand [6] The authors measured the EEG and EMG signal on 52 subjects aged 20-30 years. The EEG measurement was performed using 64 Ag/AgCl active electrodes. Since in the previous method we worked with electrodes C3, CZ and C4, for comparison purposes we also worked with these electrodes in this method. Due to computational and time requirements, we decided to work only with subject 14. This subject was selected by the authors in article [7] as one of the 5 most representative (subjects

3,14,35,43,50) These subjects achieved highest classification accuracy. The sampling frequency was 512 Hz.

III. FEATURE EXTRACTION

In our experiments we used following feature extraction methods.

A. Energy in selected frequency bands

In this method is the feature the energy in the selected frequency band is calculated as the sum of squared absolute values of spectral components [8]. The spectral components are calculated using the Fast Fourier Transform (FFT) algorithm. Also various others powers of the absolute values can be used.

B. Bispectrum and noBispectrum

One novel approach to BCI is relying on bispectrum, which is a technique developed for extracting non-linear information from non-linear signals and claiming to have good performance for motor imagery based BCI [9] The bispectrum B is relying on formula:

$$B(f_1, f_2) = X(f_1)X(f_2)X^*(f_1 + f_2) \quad (1)$$

Where f_1, f_2 are the frequencies, $X(f)$ is the Fourier spectrum of the EEG signal. So the $B(f_1, f_2)$ is two dimensional signal which can be uses as image based input feature for the classification. For performance comparison we also constructed the noBispectrum feature, contains only the $X(f_1)X(f_2)$ product, so without $X^*(f_1 + f_2)$ element. Examples of both features are given on Fig. 1.

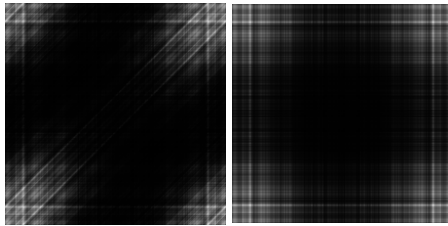


Fig. 1 Example of EEG bispectrum (left) and pure FFT outer product marked as noBispectrum (right)

C. Spectrograms and Scalograms

Spectrograms are created using the Short Time Fourier Transform (STFT) and are commonly used for BCI [10]. Scalograms are usually created using the complex continuous wavelet transform (CWT) [11]. They have exponentially varying time-frequency resolution allow variable time-frequency resolution. For our experiments, we will use wavelets Amor, Bump and Morse. We created spectrograms and scalograms using single electrode input (grayscale images) but also 3 electrode input (RGB images) where each electrode represents one RGB channel. For RGB images we have created 4 scaling levels. In the first level (LVL1), we scaled each channel in each image separately. This means that each image had its own scaling separately for all three (R, G and B) channels. In the second level (LVL2) we did separate scaling for each image separately, but not separately for each

channel. The third level (LVL3) represented scaling for each channel separately, but for all images together, and in the fourth level (LVL4) we created one global scaling for all channels and all images together.

For a single-electrode input with electrode C3, we compared whether it is appropriate to work only with the absolute value or if it will be effective if we also display the phase in the given image We created several types of adjusted spectrograms and scalograms (Fig 2), there we extend or replace the classical spectrogram/scalogram (referred as ABS) with some form which includes also phase information. Or contains zeros to be able to better compare the performance.

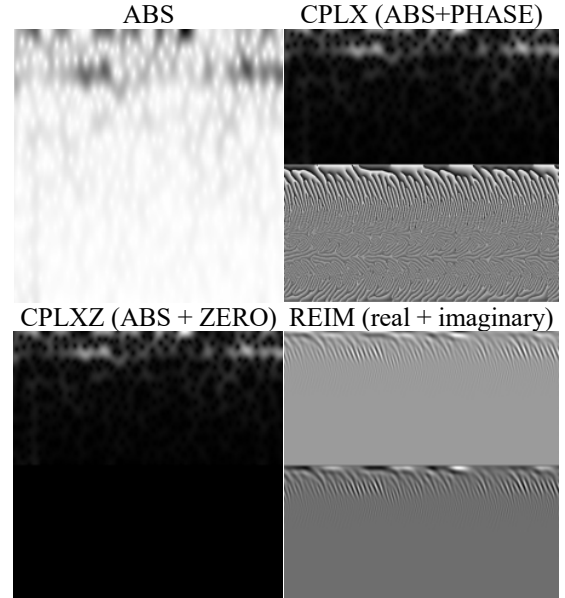


Fig. 2 Examples of time-frequency feature mappings (C3 electrode) obtained from EEG signal using continuous wavelet transform (amor wavelet), without and with the phase information in various forms. Abs means absolute value i.e. “spectrogram like” feature form. CPLX means connection of absolute value in upper part and phase in lower part of the picture. CPLXZ contains absolute value and zeroed lower half of the picture and REIM means that real and imagery parts of singal were used.

IV. CLASSIFICATION METHODS

For classification, we used simple classifiers like k-nearest neighbours (KNN) and support vector machine (SVM). We also used convolutional neural network AlexNet (pretrained on Imagenet and 1000 class recognition task and available in Matlab) but modified to comply with our number of classes.

V. EVALUATION METHODOLOGY

To achieve the most relevant results, we evaluated the obtained data using 2 types of cross-validation. In 5-fold cross-validation, we split the set of input images in the ratio 80/20. For 10-fold cross-validation, we split the input images in a ratio of 90/10. For some experiments, we used 10 repetitions of 10-fold cross validation. The random number generators were handled in that way that experiments are repeatable.

VI. EXPERIMENTS AND OBTAINED RESULTS

We performed set of experiments that using selected dataset/datasets, features and classification methods to show the classification performance based on various settings.

A. Experiment 1 – Energy + SVM and KNN classifiers

First, we searched for the optimal frequency band and determined whether it is appropriate to use an absolute value or a higher/lower power of spectral components in combination with SVM and KNN classifiers. This experiment was done in the article [4] on Dataset I and we repeated it in this paper on Dataset II. The highest accuracy is achieved when the 12 Hz frequency band is included. For both the KNN and the SVM classifiers it is ideal to use energy (second power), or the third power of the absolute value. The best results were achieved with SVM classifier (80%).

B. Experiment 2 – Scaling levels + AlexNet

In Dataset I. the highest success rate was achieved by storing LVL 1 and LVL 3. The success rate reached 91.43% when using 10-fold cross-validation [4]. In this paper, we repeated the same experiments on Dataset II. and the results showed that with Dataset II the accuracy of individual levels do not differ and the accuracy range in the interval 98.5% to 100%.

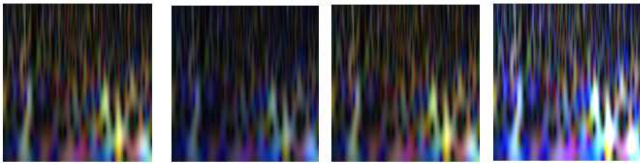


Fig 3 Examples of scalograms (Amor wavelet). From the left- LVL1, LVL2, LVL3 and LVL4

C. Experiment 4 - Phase influence + AlexNet

Influence of phase on accuracy was made only on single electrode input (C3). The main reason of this experiment was to prove that addition of phase information into input image for AlexNet has impact on resulting accuracy. The results can be seen in Fig. 4 and Fig 5. As we can see, the best results are obtained using input images which include information only about absolute value, regardless of wavelet type. The same holds also for spectrograms which performed generally worse than scalograms.

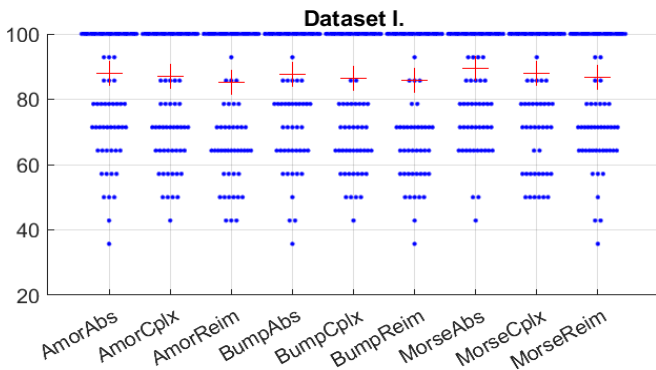


Fig 4 Detailed accuracy of 10 repetitions 10-fold cross validation – scalograms on Dataset I.

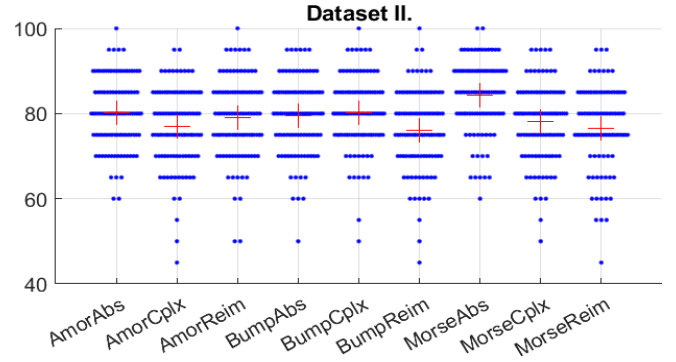


Fig 5 Detailed accuracy of 10 repetitions 10-fold crossvalidation – scalograms on Dataset 2.

D. Experiment 5 – Bispectrum and Nobispectrum + AlexNet

We created bispectrum and noBispectrum images from electrode C3 as an input to AlexNet. The results (Fig. 6, Fig.7) show, that the bispectrum and noBispectrum images both exhibit significantly worse accuracy than spectrograms or scalograms in both datasets. Even more, there seems not to be significant added value of bispectrum over noBispectrum.

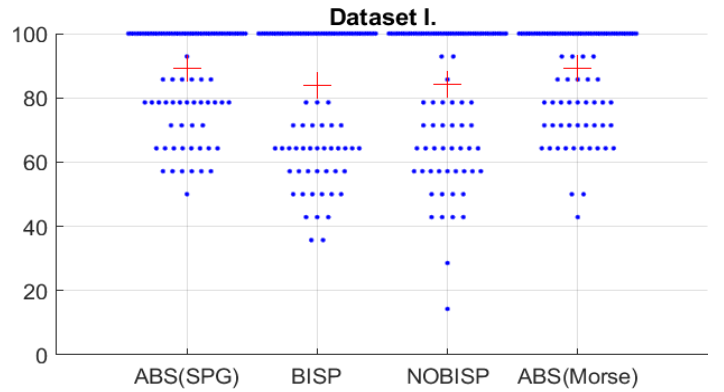


Fig 6 Detailed accuracy of 10 repetitions 10-fold crossvalidation (Dataset I.). From the left – ABS spectrogram, bispectrum, nobispectrum, ABS scalogram Morse

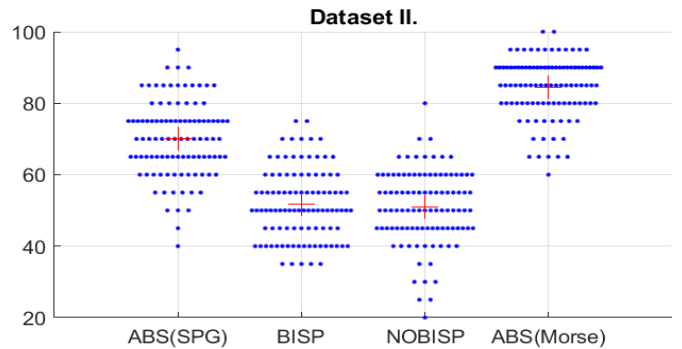


Fig 7 - Detailed accuracy of 10 repetitions 10-fold crossvalidation (Dataset II.). From the left – ABS spectrogram, bispectrum, nobispectrum, ABS scalogram Morse

E. Experiment 6 – Spectrograms and Scalograms + KNN

The authors in [11] report better performance of KNN (with low K, best performance with K=1) than AlexNet and other neural networks for image classification task for U-I diagrams. We used this KNN approach for spectrograms and scalograms

as explained in the part IV. We performed this experiment on Dataset II. Input spectrograms and scalograms were scaled from 20x20, 50x50 to 100x100 pixels. As we can see in Fig. 8, the size to which the input images were scaled did not significantly affect the KNN overall success rate. Interesting finding is that the overall performance strongly depends on type of scalogram/spectrogram used, for morse/bump-based scalograms the best performance is exhibited for K below 10 and for others the accuracy grows with K. However, no settings were found that could outperform the AlexNet.

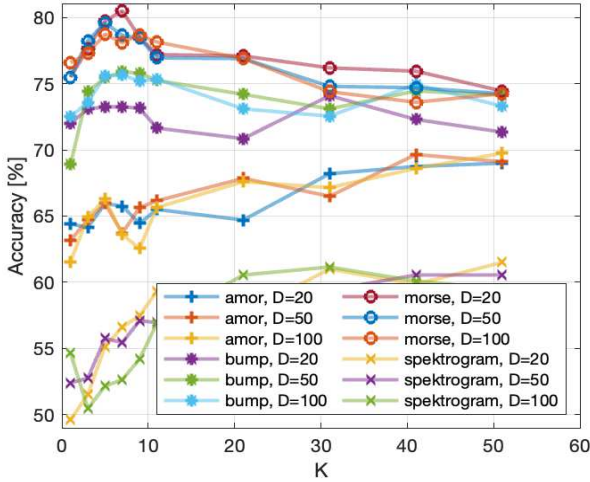


Fig 8 Comparison of different types of scalograms, spectrogram and scaling values K-NN input images, using 10 repetitions of 10-fold crossvalidation

F. Explainable

As the best results obtained so far in the paper were obtained using AlexNet, we try to understand which part of the input features (scalograms) is the generally most important part (in the time-frequency sense) of the input that contribute to the classification decision. The example results can be shown on Fig. 9 – separate for each input image, or in the form of cumulative (average) results on Fig. 10. Our experiments show that the most important part is in frequency located in the motor imagery band, but in time, it is located before the part with most energy in the band. We find this phenomenon as interesting and worth of further analysis.

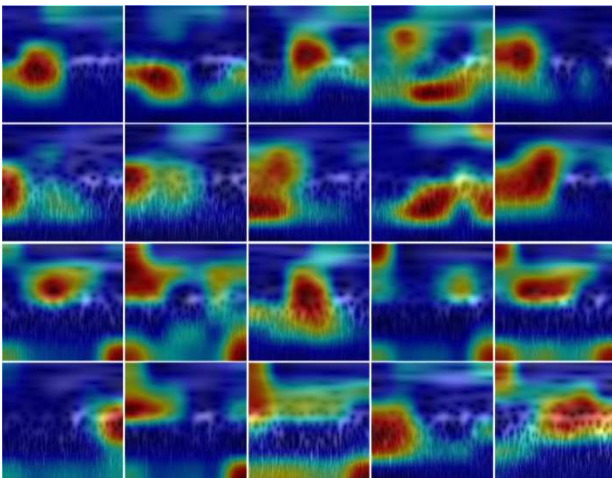


Fig. 9 Example of scalogram for all 20 images in Dataset II test set and their corresponding recognition significance for AlexNet using Grad-CAM method. Displayed Amor scalograms in grayscale overlayed with semitransparent heatmap for Grad-CAM based significance images.

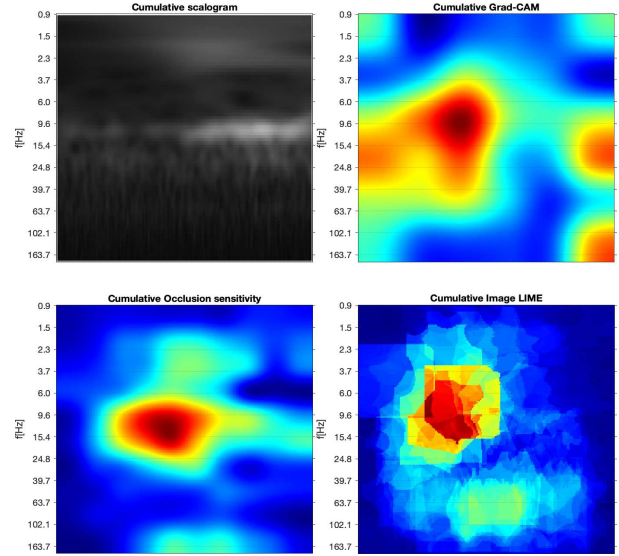


Fig. 10 Example of cumulative scalogram for all 20 images in Dataset II test set and corresponding recognition significance for Alexnet using three explainable methods

VII. CONCLUSION

In the presented contribution we focused on image-based feature representation for motor imagery BCI using neural networks. As typical example the pre-trained AlexNet was used. We did not focus to achieve the best accuracy in absolute sense (for this we would of course try also e.g., ResNet architectures and their derivatives as Inception-ResNet-v2, etc ...) but to reveal interesting properties when using various feature extraction methods coupled with selected classifiers and compare them face-to face based on the same dataset and same evaluation methodology with reasonable informative value.

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